USN | | | | | | | | | | [22ME33]

THIRD SEMESTER B. TECH EXAMINATION, DECEMBER -2023

FLUID MECHANICS AND MACHINERY

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Use of Fundamental Dimensions Table is permitted
- **iii.** Answer All Questions

		PART-A	**05 X 02=10**		
1.	**a.**	Describe the phenomenon of surface tension with an example	L2	CO1	**02**
	b.	Explain the Laminar and Turbulent flow with an example	L2	CO2	**02**
	c.	Classify different types of minor losses occurring in fluid flow	L2	CO3	**02**
	d.	Explain governing in turbine	L2	CO4	**02**
	e.	Discuss different types of head in a pump	L2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Briefly Describe the following: **i.** Surface gravity **ii.** Capillarity **iii.** Compressibility **iv.** Bulk Modulus **v.** Density	L2	CO1	**05**
		OR			
	b.	State and prove pascals law.	L2	CO1	**05**
3.	**a.**	Explain briefly the surface tension, specific weight, specific volume, cohesion and adhesion	L2	CO1	**05**
		OR			
	b.	State and prove Hydrostatic Law	L2	CO1	**05**
4.	**a.**	Distinguish between the following types of fluid flow with examples **i.** Steady and Unsteady flow **ii.** Rotational and Irrotational flow	L3	CO2	**05**
		OR			
	b.	Distinguish between the following types of fluid flow with examples **i.** Laminar and Turbulent flow **ii.** Uniform and non-uniform flows	L3	CO2	**05**
5.	**a.**	Derive Darcy-Weishbach equation in fluid flows.	L3	CO3	**05**
		OR			
	b.	Derive Buckingham's Pi-theorem.	L2	CO3	**05**
6.	**a.**	Explain the working of Pelton wheel with a neat sketch.	L2	CO4	**05**

		OR			
	b.	Explain the working of Francis turbine with a neat sketch.	L2	CO4	**05**
7.	**a.**	Explain the working of reciprocating pump with neat sketch.	L2	CO5	**05**
		OR			
	b.	Explain the working of centrifugal pump with a neat sketch.	L2	CO5	**05**
		PART-C			**06 X 10=60**
8.	**a.**	A metal plate 1.25 m × 1.25 m × 6 mm thick and weighing 90 N is placed midway in the 24 mm gap between the two vertical plane surfaces as shown in the Fig. 1.15. The gap is filled with an oil of specific gravity 0.85 and dynamic viscosity 3.0 N.s /m^2. Determine the force required to lift the plate with a constant velocity of 0.15 m/s	L3	CO1	**10**
		OR			
	b.	A pipe contains an oil of specific gravity 0.9. A differential manometer connected at the two points A and B shows a difference in mercury level as 15cm. Find the difference in pressure at the two points.	L3	CO1	**10**
9.	**a.**	Two large surfaces are 12 mm apart. This space is filled with an oil of absolute viscosity 0.972 NS/m^2. Find what force is required to drag a plate of area 0.25m^2 between the two surfaces at a speed of 0.3m/s. **i.** When the plate is equidistant from the surfaces, **ii.** when the plate is at 4mm from one of the surfaces.	L3	CO1	**10**
		OR			
	b.	A U-tube manometer is used to measure the pressure of oil of specific gravity 0.85 flowing in a pipe line. Its left end is connected to the pipe and the right-limb is open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (specific gravity = 13.6) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of the oil in the pipe.	L3	CO1	**10**
10.	**a.**	Derive the continuity equation in three dimensional Cartesian coordinates	L3	CO2	**10**
		OR			
	b.	Water is flowing through a pipe having diameters 600 mm and 400 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 350 kN/m^2 and the pressure at the upper end is 100 kN/m^2. Determine the difference in datum head if the rate of flow through the pipe is 60 litres/sec	L3	CO2	**10**
11.	**a.**	Describe Rayleigh's method to formulate a dimensionally homogeneous equation between the various physical quantities effecting a certain phenomenon.	L2	CO3	**10**

		OR			
	b.	A horizontal pipe line 40 m long is connected to a water tank at one end and discharges freely into the atmosphere at the other end. For the first 25 m of its length from the tank, the pipe is 150 mm diameter and its diameter is suddenly enlarged to 300 mm. The height of water level in the tank is 8 m above the centre of the pipe. Considering all losses of head which occur, Determine the rate of flow	L3	CO3	**10**
12.	**a.**	**i.** State causes of cavitation in hydraulic turbine and methods of avoiding it. **ii.** State advantages and disadvantages of Francis turbine over Pelton wheel.	L2	CO4	**10**
		OR			
	b.	Explain criteria for the selection of hydraulic turbines and discuss hydraulic efficiency, mechanical efficiency and overall efficiency with expression.	L2	CO4	**10**
13.	**a.**	Derive expression for the pressure rise in the impeller of a centrifugal pump by neglecting the frictional and other losses in the impeller.	L3	CO5	**10**
		OR			
	b.	Explain the following terms: Net positive suction head, priming and cavitation.	L2	CO5	**10**